

I. Project Title:

AI for Agriculture: Towards Digital Twins for Methane Emissions Forecasting and Scenario Analysis

II. Project Abstract:

Agriculture is the leading sectoral source of methane (CH₄) globally, accounting for a disproportionate share of radiative forcing given methane's global warming potential of 28× CO₂ over a 100-year horizon. Emissions from managed grassland livestock systems are projected to increase alongside rising global food demand, yet current reporting frameworks — notably IPCC Tier 1 mostly relies on static emission factors that fail to capture temporal variability, site-specific conditions, or the effects of management interventions. At the same time, existing machine learning approaches to agricultural CH₄ prediction have largely been confined to animal-scale enteric fermentation, measured via GreenFeed or respiration chambers, leaving ecosystem-scale flux prediction substantially underexplored.

This study addresses that gap by developing a digital shadow architecture for CH₄ flux forecasting at the North Wyke Farm Platform (NWFP), Devon, using half-hourly Eddy Covariance measurements from all farmlets (2018–present). To the author's knowledge, no prior study has applied machine learning forecasting to EC CH₄ flux data from a managed temperate grassland. Furthermore, translating these physical emission drivers into a predictive architecture introduces a non-trivial data fusion challenge, requiring the harmonization of continuous environmental sensors with discrete, episodic management logs.

To navigate this, the project implements a rigorous benchmarking pipeline, evaluating established tree-based ensembles against sequential deep learning architectures (e.g., LSTM, TFT) under identical experimental conditions to challenge assumptions of architectural superiority. The framework mandates an Explainable AI layer to decode multi-driver dependencies, paired with distribution-free uncertainty quantification to output statistically calibrated prediction intervals. This architecture culminates in a scenario analysis module, delivering a proof-of-concept digital shadow capable of evaluating predicted flux responses under contrasting, uncertainty-bounded management and climate conditions.

Hypotheses:

1. EC will capture a more grounded and accurate methane emissions prediction compared to Static Emission Factors (IPCC)
2. ML-based digital shadow will produce multi-step emission forecasts and statistically distinguishable, interpretable predictions under contrasting management (known) & climate (unknown) scenarios.

III. Project Aims and Objectives:

III. 1. Overall Aim:

The development and evaluation of multiple ML approaches for CH₄ forecasting at North Wyke Farm Platform and demonstrate its integration with a digital shadow architecture with scenario analysis and uncertainty quantification.

III. 2. Specific Objectives:

1. Conduct a systematic literature review on time-series forecasting, quantile machine learning, eddy covariance sensor methodology, and agricultural digital twin frameworks.
2. Acquire, pre-process, and rigorously document NWFP EC CH₄ GHG flux data (2018–present), including gap imputation, quality control, and feature engineering from complementary meteorological and soil variables.
3. Design and implement a benchmarking pipeline comparing LSTM, Temporal Fusion Transformer, Random Forest, and XGBoost models against persistence and seasonal mean baselines, using temporal cross-validation to prevent data leakage.
4. Apply interpretability methods (permutation importance, SHAP) and uncertainty quantification via quantile machine learning or conformal prediction to characterise model behaviour and produce calibrated prediction intervals.
5. Generate synthetic management and climate scenarios and evaluate in-sample and out-of-sample model performance under contrasting conditions.
6. Develop a digital shadow interface (Streamlit) integrating the forecasting model with scenario analysis and uncertainty visualisation for farm management support.

IV. Planning Review Protocol: Research Questions (CWA-T2) and Development (CWA-T3)

IV. 1. Initial P-C-S Framing (CWA-T2):

- **Problem:**

Ecosystem-scale CH₄ flux forecasting from EC data at a managed temperate grassland livestock farm (NWFP), where emissions are driven by multiple interacting factors beyond enteric fermentation alone.

- **Constraint:**

EC CH₄ data has not been applied to ML forecasting in this context; existing ML work on EC CH₄ is confined to gap-filling. Beyond this, the forecasting pipeline must integrate multiple heterogeneous data streams such as soil moisture, temperature, rainfall, management activities, each with differing temporal granularities and gaps in the data. This creates a non-trivial data harmonization and imputation challenge that further complicates model selection and makes interpretability essential.

- **Solution:**

A comparative ML forecasting pipeline (statistical baselines through deep learning) combined with interpretability, uncertainty quantification, and a scenario analysis module, forming a digital shadow for emissions understanding under varying management conditions.

IV. 2. Research Questions (CWA-T2):

RQ1: What ML approaches have been applied to CH₄ flux prediction in agricultural and grassland contexts, do any address ecosystem-scale EC data, and how does the transition from within-distribution gap-filling to multi-step temporal forecasting alter the modelling constraints?

RQ2: How do statistical baselines, tree-based ensemble methods, and deep learning architectures compare in forecasting half-hourly EC CH₄ flux under conditions of temporal variability, limited prior benchmarking, and non-stationary nature of managed grasslands?

RQ3: What is the strength of the evidence supporting these approaches, and how transferable are findings from adjacent domains (wetland gap-filling, animal-scale prediction) to managed grassland EC forecasting?

RQ4: How can Explainable AI (XAI) and rigorous uncertainty quantification (e.g., Quantile Machine Learning) be utilized to decode the complex interactions between continuous environmental variables and discrete management interventions?

RQ5: What are the structural and methodological requirements to integrate these predictive models to a functional digital shadow architecture that demonstrates “what-if” scenario analysis at a farm-scale managed grasslands?

IV. 3. Research Questions Development (CWa-T3):

- Initial Formulation:

The initial research question was broadly framed as: *"What ML approaches exist for predicting agricultural CH₄ emissions, and how do they compare?"* The problem was understood in general terms, the constraint loosely defined as data availability and model selection, and the solution described broadly as a "digital twin with forecasting capabilities". No distinction was made between measurement paradigms, model classes, or data integration challenges.

- Critical Evaluation After Preliminary Screening:

Problem: The NWFP holds over seven years of continuous half-hourly EC measurements that have never been applied to ML-based temporal forecasting. The most relevant existing work at NWFP, Partridge et al. (2024), relies on GreenFeed which captures intermittent animal-scale breath samples — incompatible with the integrated ecosystem-scale signal that EC captures across enteric fermentation, soil processes, and manure simultaneously.

Constraint: Contradictory model performance evidence across adjacent domains makes a systematic benchmark genuinely necessary rather than assumed. Beyond this, EC-based forecasting introduces complex driver interactions, data streams at differing temporal granularities, and persistent sensor gaps, each of which complicates model selection and makes interpretability essential.

Solution: Purcell & Neubauer (2023) and Fakeye et al. (2024) together establish that what-if scenario simulation remains critically absent from agricultural digital twin implementations despite its significant potential for farm management decision support.

- Refinement and Justification:

The initial framing was refined across all three P-C-S components. The problem was narrowed to ecosystem-scale EC forecasting at managed temperate grassland, distinguishing the NWFP EC signal from animal-scale paradigms. The constraints were expanded to model selection uncertainty under contradictory benchmark evidence and multi-source data heterogeneity. The solution was repositioned from a generic digital twin to a digital shadow with explicit scenario analysis capability. The final five RQs are defensible because each refinement is confirmed by multiple independent sources rather than assumed in advance.

V. Search Results and Study Selection (Cwa-T7) and (Cwa-T8)

V. 1. Implementing Search Strategy

A search on all of the keywords from Table 3 has been conducted, yielding 768 results. Due to the high volume of papers, we further collapse the results into 3 categories.

- Group 1: ML / AI Methods – covers papers related to the methodology of the project such as time series architectures, forecasting benchmarks, uncertainty quantification, and interpretability.
- Group 2: Emissions and Measurement Methodology – covers papers related to the prediction target and measurement context of the project such as Eddy Covariance CH₄ flux prediction, grassland GHG modelling, enteric fermentation drivers, etc.
- Group 3: Application and Deployment Context – covers papers related to the exact environment of Rothamsted research’s North Wyke Farm Platform and the infrastructure . This category covers NWFP specific publications such as agricultural digital twin frameworks, multi-source data fusion pipelines in agricultural settings, and the digital infrastructure where this project operates on.

From these three groups, by combining all search terms with (“AND”), we can retrieve **44** articles assessed for eligibility. A diagram of these three groups and their overlap can be seen in Figure 1.

V. 2. Study Selection

Following initial title and abstract screening, the supervisor (Prof. Harris) was consulted to identify studies not captured by the database search strategy. This resulted in the addition of six studies: one institutional data guide providing technical documentation of the NWFP sensor infrastructure and flux processing pipeline; one EC flux gap-filling study (Zhu et al.); one EC flux measurement study at NWFP (Cardenas et al.); one farm-scale digital twin framework paper (Fakeye et al); one IPCC emissions reference; and one quantile machine learning study (Oulaid et al.).

These additions reflect two refinements to the protocol. First, grey literature and institutional reports from Rothamsted Research are insufficiently indexed by Scopus and arXiv, meaning the data guide would not have surfaced through the keyword strategy alone. Second, the supervisor identified domain-critical papers at the intersection of UQ methodology and site-specific EC measurement that the search queries, which were optimised for breadth, had not ranked highly enough for inclusion.

Two justifications underpin these additions. First, institutional data guides and grey literature from Rothamsted Research are filtered out through keyword searches and publication type filter. Second, the supervisor identified domain-critical papers specific to NWFP and Rothamsted Research that the broad search strategy had not captured. No papers were excluded following supervisor consultation. These six additions brought the final eligible study count to n=29, as summarised in Figure 2.

VI. Background and Critical Evaluation

Introduction

Agricultural greenhouse gas emissions represent one of the most intractable challenges in climate mitigation. Global livestock emissions are projected to rise from 6.2 to 9.1 GtCO₂eq by 2050 driven by increasing demand for animal products [1], with methane (CH₄) being approximately 80 times more potent than CO₂ over a 20-year horizon [2] and accounting for roughly 49% of all agricultural greenhouse gas emissions through enteric fermentation alone [3]. The North Wyke Farm Platform (NWFP) offers a rare and underutilized resource to address this: continuous half-hourly eddy covariance (EC) flux measurements from managed beef cattle grassland spanning over seven years, capturing the integrated ecosystem-scale CH₄ signal from enteric fermentation, soil processes, and manure simultaneously [14], [25].

Despite this, to the author's knowledge, no study has applied ML-based temporal forecasting to NWFP EC CH₄ data, and no system exists that integrates this signal with scenario analysis capabilities for farm management decisions. Addressing this gap requires not only a forecasting model but a deployment architecture capable of what-if simulations, a capability identified as critically absent from agricultural digital twin systems. This review synthesizes evidence from 29 eligible studies across five domains to establish that the NWFP CH₄ forecasting problem is a structurally distinct, underserved research target, and that the proposed methodology constitutes the necessary and appropriate response.

Agricultural Digital Twins: Importance and Difficulty of Research

Agricultural digital twins are genuinely difficult to build, and understanding why is essential for positioning this research correctly. Unlike industrial systems where the physical environment is controlled and deterministic, agricultural systems are living systems and are biologically variable, weather-dependent, and non-stationary by nature [5]. Fakeye et al. [4] identified three compounding challenges that any farm-scale DT must simultaneously resolve: model integration, where neither process-based models (PBMs) nor data-driven models (DDMs) are sufficient alone — PBMs fail to capture natural system variability while DDMs are susceptible to overfitting and remain inherently black-box; data infrastructure, where sensor malfunctions, missing values, and legacy systems not designed for real-time dynamic flows create persistent data quality problems; and software engineering, where scalability, computational efficiency, and interoperability require Kubernetes orchestration and HPC infrastructure simply to operate. The result is that while the theoretical promise of an agricultural DT relies on seamless, bidirectional integration of sensing and physical automation, current implementations predominantly function as rare unidirectional digital shadows, with true bidirectional twins rarer still [4], [24].

The critical limitation of current systems is not observation but what comes next. Purcell and Neubauer [5], [6] both identify through systematic review that what-if scenario simulation, the defining capability that would transform a DT from a monitoring tool into a decision-support system remains critically absent from agricultural DT applications. Here, NWFP represents an opportunity, as it has already solved the instrumentation problem: over seven years of 15-minute to 30-minute multi-sensor data across soil, water, atmosphere, and livestock sub-systems provides the data foundation that most farms cannot offer, and Fakeye et al. [4] have proposed the three-tier architecture and identified GHG emissions as a named target. What does not yet exist is the CH₄ forecasting component with scenario analysis capability. To advance beyond mere observation, a predictive pipeline must explicitly integrate uncertainty-bounded forecasting

with dynamic scenario analysis that bridges the gap between a passive digital shadow and a functional, decision-enabling twin.

EC-ML for CH₄ Prediction: Benchmarks, Limits, and Absence of Managed Grassland Scenario

The EC-ML literature for CH₄ flux prediction is dominated by tree-based ensemble methods, and this dominance is well-evidenced but narrowly scoped [19], [20]. Irvin et al. [7] established the benchmark across 17 FLUXNET-CH₄ wetland sites, finding RF and XGBoost superior to Artificial Neural Networks (ANN) and penalised linear regression across all biomes, with soil temperature as the most important predictor. Kim et al. [8] confirmed RF dominance across five wetland and rice paddy sites, adding an important feature engineering insight: lagged environmental variables improve CH₄ prediction substantially more than CO₂, reflecting the temporally complex and nonlinear nature of methane emission. Zhu et al. [9] extended these findings to challenging ecosystems including UK managed pastures, validating RF-based gap-filling as the recommended approach for the precise ecosystem type studied at NWFP.

While consistent, these benchmarks exclusively address gap-filling, a within-distribution interpolation task, rather than multi-step temporal forecasting. The distinction matters as gap-filling asks a model to reconstruct missing observations within a known regime; forecasting asks it to extrapolate sequentially into future states, where temporal dependencies become the dominant modelling challenge. Existing benchmarks do not answer the forecasting question.

A further limitation runs deeper than task type. Irvin et al. [7] documented that raw ML uncertainty estimates were systematically underestimated across all 17 benchmark sites, with calibration corrections required to produce reliable annual budget estimates. This finding is not incidental as it signals that point prediction accuracy and uncertainty reliability are separable properties, and that a model which performs well on RMSE may still produce misleading confidence about its outputs. For a project that proposes scenario analysis as a deliverable, this distinction is noteworthy: scenario predictions without calibrated uncertainty intervals are not actionable.

The application of deep learning architectures to EC CH₄ flux prediction remains narrowly scoped. While LSTM-based approaches have emerged in wetland upscaling literature [21], to this author's knowledge, no peer-reviewed study has applied a Temporal Fusion Transformer or attention-based Transformer architecture to EC CH₄ flux on any ecosystem type. Furthermore, while basic Artificial Neural Networks (ANNs) have been tested for half-hourly gap-filling in wetlands and rice paddies [16], [17], [23], advanced sequential deep learning architectures remain entirely absent from ecosystem-scale forecasting in managed agricultural grasslands.

More fundamentally, the comparative question of whether temporal architectures outperform tree-based methods for CH₄ forecasting at farm scale remains entirely open. Critically, the wider time series literature cautions against assuming the answer. Zeng et al. [10] demonstrated across nine standard benchmarks that simple linear models outperform transformer architectures, attributing the failure to permutation-invariant self-attention destroying temporal ordering. This finding does not disqualify deep learning for EC CH₄, but it does make the case against assuming architectural superiority in advance. The appropriate response is therefore to benchmark statistical baselines, RF, XGBoost, LSTM, and TFT under identical experimental conditions and let the evidence decide. One notable existing work on emission prediction on NWFP, Partridge et al. [11], achieves $r=0.619$ on NWFP cattle CH₄ with Gradient Boosting, demonstrates that ML

is applicable to NWFP data and that substantial room for improvement remains. However, the animal-scale GreenFeed measurement context, wherein intermittent breath samples from individual animals [31] are fundamentally distinct from continuous ecosystem-scale EC flux means that the EC forecasting question remains unanswered.

CH₄ Emission Drivers and Multi-Source Data Fusion Challenges

Accurate ecosystem-scale CH₄ forecasting in managed UK grasslands requires models capable of capturing complex, multi-driver dependencies. In these settings, emissions are not driven by a single variable, but by the continuous interaction of biophysical constraints such as soil moisture, rainfall, temperature and even management practices like grazing schedules and fertilizer additions [26], [27], [28], [29]. While tree-based algorithms have proven adept at modelling these non-linear relationships, Buzacott et al. [12] demonstrated through the application of Shapley values that raw predictive accuracy often masks a highly entangled multi-driver dependency [12], [22]. Furthermore, Sharma et al. [13] cautioned that standalone machine learning models lacking site-specific management calibration frequently miss episodic, high-emission events. In a managed grassland context, ignoring discrete management interventions guarantees that peak emission fluxes will be under-predicted [30].

Translating these driver dependencies into a predictive architecture is a non-trivial data fusion challenge. The necessary features exist on different and sometimes incompatible temporal scales: eddy covariance and meteorological sensors generates time series data with different intervals, whereas management logs are sparse, discrete events. Recent work at the NWFP provides a blueprint for managing this heterogeneity. Partridge et al. [11] successfully fused disparate farm environment records with animal-scale measurements, utilizing gradient boosting and Explainable AI (XAI) to map emission drivers. However, while this demonstrates the viability of multi-source fusion at NWFP, Partridge et al.'s reliance on the GreenFeed system renders the approach fundamentally incompatible with ecosystem-scale forecasting as it captures intermittent, animal-specific breath samples only.

Addressing this gap requires an architecture that can not only fuse continuous and discrete data streams but also handle the inevitable gaps in EC sensor data. In this case, Zhu et al. [9], [33] demonstrated that machine learning approaches (specifically Random Forest) can successfully extrapolate and gap-fill missing EC fluxes over challenging managed pastures, validating the use of substitute reanalysis data (like ERA5) when local sensors fail. Ultimately, an effective digital shadow must harmonize these multi-resolution data inputs, bridge sensor gaps, and mandate an interpretability layer to decode how interacting discrete and continuous drivers influence the final CH₄ forecast [14], [15].

Synthesis: Refining Project Assumptions

The synthesized evidence critically interrogates trends in environmental machine learning, ultimately validating and reinforcing the project's initial methodological framing. While recent literature often assumes that advanced deep learning temporal architectures (such as TFT or LSTM) inherently outperform established tree-based methods for sequence prediction, the wider literature [7], [8], [9], [21], [22] fundamentally challenges this assumption. Zeng et al. [10] even demonstrated across multiple rigorous benchmarks that simple linear models frequently outperform complex Transformer architectures, attributing this to permutation-invariant self-attention mechanisms destroying temporal ordering. This finding confirms the project's foundational hypothesis: architectural superiority cannot be assumed. It

directly justifies the project's requirement for a rigorous benchmarking pipeline that forces statistical baselines, tree-based ensembles, and deep learning models to compete under identical experimental conditions.

Furthermore, the literature exposes the severe limitations of evaluating agricultural models solely on point-prediction accuracy. Both Irvin et al. [7] and Kim et al. [8] highlighted that CH₄ fluxes are highly non-stationary and prone to extreme episodic peaks, resulting in raw ML uncertainty estimates that are systematically underestimated. In the context of decision support, a model that performs well on average but fails to quantify its uncertainty during extreme events is actively dangerous for scenario planning. This evidence validates the project's pre-existing mandate to integrate rigorous uncertainty quantification. The literature confirms that reliable, calibrated prediction intervals achieved through distribution-free methods like conformal predictions and quantile machine learning is a structural requirement for generating actionable "what-if" scenarios.

VII. Conclusions and Methodological Recommendations

The ecosystem-scale prediction of CH₄ in managed temperate grasslands remains a critical blind spot in current ML-driven agricultural literature. To address this and construct a functional digital shadow for NWFP, this review recommends the following pipeline:

1. **Rigorous Benchmarking:** Evaluate RF and XGBoost against sequential architectures (LSTM, TFT) and persistence baselines, using strict temporal cross-validation to prevent leakage — motivated by the gap-filling dominance of tree-based methods [7], [8], [9] and the caution against assumed transformer superiority [10].
2. **Heterogeneous Data Handling:** Engineer a feature space harmonising continuous meteorological and EC variables with discrete management logs, using ERA5 reanalysis substitution [9] and field-scale extrapolation methods [33] to bridge sensor gaps.
3. **Interpretability and Calibration:** Apply SHAP-based XAI [11], [12], [22] to decode multi-driver interactions, paired with quantile ML or conformal prediction [15] to output calibrated prediction intervals, given evidence that uncalibrated uncertainty is a systematic failure mode in EC contexts [7], [18].

By implementing this methodology, the project will advance the state-of-the-art from passive observation to predictive scenario analysis, delivering a proof-of-concept digital shadow capable of informing sustainable agricultural policy and farm-level interventions.

References

- [1] FAO, *Methane emissions in livestock and rice systems – Sources, quantification, mitigation and metrics*, Rome, Italy: Food and Agriculture Organization of the United Nations, 2023. <https://doi.org/10.4060/cc7607en>
- [2] IPCC, *Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*, V. Masson-Delmotte *et al.*, Eds. Cambridge, UK: Cambridge University Press, 2021.
- [3] EEA, *Greenhouse gas emissions from agriculture in Europe*, European Environment Agency, 2024.
- [4] I. Fakeye *et al.*, "Towards A Framework For Farm-scale Digital Twins," in Proc. MODELS 2024 - ACM/IEEE 27th Int. Conf. on Model Driven Engineering Languages and Systems, 2024, pp. 486–491. <https://doi.org/10.1145/3652620.3688264>
- [5] W. Purcell and T. Neubauer, "Digital Twins in Agriculture: A State-of-the-art review," *Smart Agricultural Technology*, vol. 3, 2023. <https://doi.org/10.1016/j.atech.2022.100094>
- [6] W. Purcell, T. Neubauer, and K. Mallinger, "Digital Twins in agriculture: challenges and opportunities for environmental sustainability," *Current Opinion in Environmental Sustainability*, vol. 61, 2023. <https://doi.org/10.1016/j.cosust.2022.101252>
- [7] J. Irvin *et al.*, "Gap-filling eddy covariance methane fluxes: Comparison of machine learning model predictions and uncertainties at FLUXNET-CH₄ wetlands," *Agricultural and Forest Meteorology*, vol. 308-309, p. 108528, 2021. <https://doi.org/10.1016/j.agrformet.2021.108528>
- [8] Y. Kim *et al.*, "Gap-filling approaches for eddy covariance methane fluxes: A comparison of three machine learning algorithms and a traditional method with principal component analysis," *Global Change Biology*, vol. 26, no. 3, pp. 1499–1518, 2020. <https://doi.org/10.1111/gcb.14845>
- [9] S. Zhu *et al.*, "Gap-filling carbon dioxide, water, energy, and methane fluxes in challenging ecosystems: Comparing between methods, drivers, and gap-lengths," *Agricultural and Forest Meteorology*, vol. 332, p. 109365, 2023. <https://doi.org/10.1016/j.agrformet.2023.109365>
- [10] A. Zeng, M. Chen, L. Zhang, and Q. Xu, "Are Transformers Effective for Time Series Forecasting?", *AAAI Conference on Artificial Intelligence*, 2023. www.aaai.org
- [11] T. Partridge, B. Li, B. Alhnaity, and Q. Meng, "Utilizing Machine Learning to Understand and Predict Methane Emissions in Cattle Farming with Farm-Scale Environmental and Biological Variables," in Proc. ICCA 2024, 2024. <https://doi.org/10.1109/ICCA62237.2024.10927962>
- [12] A. J. V. Buzacott *et al.*, "Drivers and Annual Totals of Methane Emissions From Dutch Peatlands," *Global Change Biology*, vol. 30, no. 12, 2024. <https://doi.org/10.1111/gcb.17590>
- [13] P. Sharma *et al.*, "Coupled machine learning–ecosystem ensemble models substantially improve predictions of nitrous oxide (N₂O) fluxes from US croplands," *Proceedings of the National Academy of Sciences (PNAS)*, vol. 123, no. 10, 2026. <https://doi.org/10.1073/pnas.2524808123>
- [14] J. M. B. Hawkins, H. Sint, and P. Harris, *The North Wyke Farm Platform: Data Guide*, Version 1.0. Harpenden, UK: Rothamsted Research, 2025. <https://doi.org/10.23637/rothamsted.99440>

- [15] B. Oulaid *et al.*, "Geographically Weighted Quantile Machine Learning for Probabilistic Soil Moisture Prediction from Spatially Resolved Remote Sensing," *Remote Sensing*, vol. 17, no. 16, 2907, 2025. <https://doi.org/10.3390/rs17162907>
- [16] S. Kheradmand, N. Heidarzadeh, and S. H. Kia, "Prediction of the CO₂ emission across grassland and cropland using tower-based eddy covariance flux measurements: a machine learning approach," *Environment, Development and Sustainability*, vol. 25, no. 6, pp. 5495–5509, 2023. <https://doi.org/10.1007/s10668-022-02276-9>
- [17] L. Bigaignon, R. Fieuzal, C. Delon, and T. Tallec, "Combination of two methodologies, artificial neural network and linear interpolation, to gap-fill daily nitrous oxide flux measurements," *Agricultural and Forest Meteorology*, vol. 291, p. 108037, 2020. <https://doi.org/10.1016/j.agrformet.2020.108037>
- [18] J. P. Goodrich *et al.*, "Improved gap filling approach and uncertainty estimation for eddy covariance N₂O fluxes," *Agricultural and Forest Meteorology*, vol. 297, p. 108280, 2021. <https://doi.org/10.1016/j.agrformet.2020.108280>
- [19] M. Jung *et al.*, "Scaling carbon fluxes from eddy covariance sites to globe: Synthesis and evaluation of the FLUXCOM approach," *Biogeosciences*, vol. 17, no. 5, pp. 1343–1365, 2020. <https://doi.org/10.5194/bg-17-1343-2020>
- [20] W. Liang *et al.*, "Estimation of Global Grassland Net Ecosystem Carbon Exchange Using a Model Tree Ensemble Approach," *Journal of Geophysical Research: Biogeosciences*, vol. 125, no. 1, 2020. <https://doi.org/10.1029/2019JG005034>
- [21] G. McNicol *et al.*, "Upscaling Wetland Methane Emissions From the FLUXNET-CH₄ Eddy Covariance Network (UpCH₄ v1.0): Model Development, Network Assessment, and Budget Comparison," *AGU Advances*, vol. 4, no. 5, 2023. <https://doi.org/10.1029/2023AV000956>
- [22] Z. Ouyang *et al.*, "Paddy rice methane emissions across Monsoon Asia," *Remote Sensing of Environment*, vol. 284, p. 113335, 2023. <https://doi.org/10.1016/j.rse.2022.113335>
- [23] L. Ma *et al.*, "Improving gap-filling performance for CH₄ fluxes of eddy covariance data by combining marginal distribution sampling and machine learning algorithm over paddy fields," *Journal of Cleaner Production*, vol. 510, 2025. <https://doi.org/10.1016/j.jclepro.2025.145615>
- [24] S. Mishra and S. K. Sharma, "Advanced contribution of IoT in agricultural production for the development of smart livestock environments," *Internet of Things*, vol. 22, 100724, 2023. <https://doi.org/10.1016/j.iot.2023.100724>
- [25] L. M. Cardenas *et al.*, "CO₂ fluxes from three different temperate grazed pastures using Eddy covariance measurements," *Science of the Total Environment*, vol. 831, 154819, 2022. <https://doi.org/10.1016/j.scitotenv.2022.154819>
- [26] I. Feigenwinter, L. Hörtnagl, and N. Buchmann, "N₂O and CH₄ fluxes from intensively managed grassland: The importance of biological and environmental drivers vs. management," *Science of the Total Environment*, vol. 903, 166389, 2023. <https://doi.org/10.1016/j.scitotenv.2023.166389>

- [27] L. Barczyk, J. Six, and C. Ammann, "Partitioning and driver analysis of eddy covariance derived N₂O emissions from a grazed and fertilized pasture," *Agricultural and Forest Meteorology*, vol. 359, 110278, 2024. <https://doi.org/10.1016/j.agrformet.2024.110278>
- [28] S. K. Jones *et al.*, "The nitrogen, carbon and greenhouse gas budget of a grazed, cut and fertilised temperate grassland," *Biogeosciences*, vol. 14, no. 8, pp. 2069–2088, 2017. <https://doi.org/10.5194/bg-14-2069-2017>
- [29] S. H. Knox *et al.*, "Biophysical controls on interannual variability in ecosystem-scale CO₂ and CH₄ exchange in a California rice paddy," *Journal of Geophysical Research: Biogeosciences*, vol. 121, no. 3, pp. 978–1001, 2016. <https://doi.org/10.1002/2015JG003247>
- [30] K. Fuchs *et al.*, "Multimodel Evaluation of Nitrous Oxide Emissions From an Intensively Managed Grassland," *Journal of Geophysical Research: Biogeosciences*, vol. 125, no. 1, 2020. <https://doi.org/10.1029/2019JG005261>
- [31] L. da Silveira Pontes *et al.*, "Performance and methane emissions by beef heifer grazing in temperate pastures and in integrated crop-livestock systems: The effect of shade and nitrogen fertilization," *Agriculture, Ecosystems and Environment*, vol. 253, pp. 90–97, 2018. <https://doi.org/10.1016/j.agee.2017.11.009>
- [32] S. Feron *et al.*, "Recent increases in annual, seasonal, and extreme methane fluxes driven by changes in climate and vegetation in boreal and temperate wetland ecosystems," *Global Change Biology*, vol. 30, no. 1, 2024. <https://doi.org/10.1111/gcb.17131>
- [33] S. Zhu *et al.*, "Eddy covariance fluxes over managed ecosystems extrapolated to field scales at fine spatial resolutions," *Agricultural and Forest Meteorology*, vol. 342, 109675, 2023. <https://doi.org/10.1016/j.agrformet.2023.109675>

APPENDIX A. Sources Documentation Table

Table 1: Sources Documentation

Data Source	Type	Date of Search	Years Covered	Justification
Scopus	Digital Library (https://www.scopus.com)	06 Apr 2026	2015–2026	<u>Primary database</u> providing comprehensive cross-disciplinary indexing of peer-reviewed journals in ML, environmental science, and agriculture (incl. IEEE, ACM, Elsevier, Springer). Covers all major outlets relevant to this project; Web of Science and IEEE Xplore excluded due to overlap.
arXiv	Preprint Server (https://arxiv.org)	06 Apr 2026	2020–2026	<u>Targeted preprint</u> server for ML methodology only (TFT, LSTM, conformal prediction, QML). Justified as cutting-edge ML methods circulate here 12–18 months before journal publication; not used for domain literature.
Rothamsted Research Repository	Institutional Repository (https://repository.rothamsted.ac.uk)	20 Mar 2026	2020–2026	<u>Institutional repository</u> providing exclusive access to NWFP technical reports, working papers, and internal datasets (e.g. Fakeye et al., 2024) not captured by any external database. Searched manually by author and project name.

APPENDIX B. Search Strategy Keywords Table (CWa-T4)

Keywords are organised into five groups reflecting the P–C–S framing. The OR operator is applied within groups to capture synonymous terms; the AND operator is applied between groups to identify studies at the intersection of the research areas.

Table 2: Search Strategy Keywords

Term / Concept	Synonymous Keywords (joined by OR within group)
Machine Learning / AI Methods	machine learning, deep learning, neural network, random forest, XGBoost, gradient boosting, LSTM, long short-term memory, transformer, temporal fusion transformer, attention mechanism, ensemble, support vector machine
Target Gas / Emission Variable	methane, CH ₄ , greenhouse gas, GHG, trace gas, emission
Measurement Technique	eddy covariance, flux tower, gap-filling, GreenFeed, respiration chamber, half-hourly flux
Agricultural / Grassland Context	grassland, pasture, grazed, livestock, temperate, managed grassland, beef cattle, dairy, enteric fermentation, ruminant, farm-scale, agriculture
Time Series Forecasting	time series forecasting, sequence prediction, temporal modelling, multivariate forecasting, autoregression, uncertainty quantification, probabilistic forecasting
Digital Twin / Smart Farming	digital twin, digital shadow, farm digital twin, smart farming, precision agriculture, scenario simulation, what-if analysis
Data Integration / Fusion	data fusion, multi-source data, heterogeneous data, multi-granularity, temporal resolution, data imputation, missing data, feature engineering, multi-modal, sensor integration

APPENDIX C. Search Strategy Table (CWa-T4)

The queries below implement the keyword strategy across the selected sources. Queries progress from broad (Q1–Q2) to targeted intersections (Q3–Q14), establishing baseline counts for the Venn diagram before narrowing to core eligible studies. The Results column is populated during the conducting phase (CWa-T7).

Table 3: Search Queries and Results

ID	Source	Search Query	Justification	Results
Q1	Scopus	("eddy covariance") AND ("methane" OR "CH4") AND ("machine learning" OR "neural network" OR "random forest")	Core intersection: EC flux measurement technique with ML applied to CH4. Targets Area 1 directly.	49
Q2	Scopus	("greenhouse gas" OR "GHG") AND ("flux" OR "emission") AND ("random forest" OR "XGBoost" OR "gradient boosting") AND ("grassland" OR "pasture")	Extends to grassland-specific GHG prediction with tree-based methods — directly tests dissertation baseline models.	35
Q3	Rothamsted Repository	"eddy covariance" OR "GHG flux" OR "methane" OR "North Wyke"	NWFP-specific search to capture all Rothamsted EC and GHG publications, including grey literature.	68
Q4	arXiv	("time series forecasting") AND ("transformer" OR "attention") AND ("multivariate")	Finds recent preprints applying attention-based forecasting to environmental data — fills post-2023 publication gap.	247
Q5	Scopus	("temporal fusion transformer" OR "LSTM") AND ("time series") AND ("forecasting") AND ("benchmark" OR "comparison") AND ("agriculture" OR "livestock")	Identifies comparative studies evaluating forecasting architectures, essential for methodological triangulation.	191
Q6	Scopus	("DLinear" OR "NLinear" OR "linear model") AND ("time series") AND ("transformer") AND ("comparison")	Specifically targets the transformer vs. linear model debate (Zeng et al. 2023 line of work).	8
Q7	Scopus	("digital twin") AND ("agriculture" OR "farming" OR "farm-scale") AND ("machine learning" OR "AI") AND ("grassland" OR "livestock")	Primary sweep for agricultural DT literature applying ML. Necessary to establish maturity baseline.	42
Q8	Scopus	("enteric fermentation" OR "rumen") AND ("methane") AND ("grassland" OR "livestock" OR "beef cattle") AND ("temperate")	Targets enteric CH4 driver studies in temperate livestock systems matching the NWFP context.	15
Q9	Scopus	("grassland") AND ("methane") AND ("eddy covariance") AND ("driver" OR "variability" OR "seasonal")	Identifies EC studies characterising environmental and management drivers of grassland GHG variability.	11
Q10	Scopus	("data fusion" OR "sensor fusion") AND ("heterogeneous" OR "multi-source") AND ("time series") AND ("agriculture" OR "environmental") AND ("machine learning" OR "AI" OR "Artificial intelligence")	Targets methods for integrating multi-resolution data streams — core challenge of the dissertation pipeline.	20

COMP0190: Project Preparation Module
Structured Literature Review Coursework

ID	Source	Search Query	Justification	Results
Q11	arXiv	("temporal fusion transformer") AND ("static covariate" OR "known future" OR "heterogeneous input")	Finds application and analysis papers exploiting TFT multi-input architecture for mixed-granularity data streams.	10
Q12	Scopus	("conformal prediction" OR "uncertainty quantification" OR "prediction interval" OR "Monte Carlo dropout" OR "deep ensemble" OR "quantile machine learning") AND ("time series" OR "forecasting") AND ("flux" OR "eddy covariance") AND ("ML" OR "AI")	Uncertainty Quantification for ML/time series in context of eddy covariance and agricultural settings	65
Q13	Scopus	("feature importance" OR "variable importance" OR "interpretability" OR "explainability" OR "SHAP") AND ("machine learning" OR "random forest" OR "gradient boosting") AND ("greenhouse gas" OR "flux" OR "eddy covariance") AND ("agricultural" OR "grassland")	Interpretability / Feature Importance for Environmental ML	7

APPENDIX D. Inclusion and Exclusion Criteria Table (CWa-T5)

Table 4: Selection Criteria

Type	Criterion ID	Criterion	Justification
Inclusion	I1	Published 2015–2026 (exceptions granted for seminal pre-2015 architecture papers: LSTM foundations, N-BEATS, and established EC gap-filling methodology)	Ensures recency and relevance to current model capabilities, while retaining foundational architectural references central to the dissertation's model comparison.
Inclusion	I2	Peer-reviewed journal articles or conference proceedings (example: IEEE, NeurIPS, ICLR)	Ensures scientific credibility and reproducibility. Conference proceedings are included for major ML architecture advances
Inclusion	I3	Written in English	Language of primary scientific literature in the target domains; ensures comprehensive interpretation without translation bias.
Inclusion	I4	Reports quantitative evaluation metrics (at minimum: RMSE, MAE, R^2 , or equivalent); allows comparison across included studies	Required for evidence synthesis and for populating Table 6 with comparable performance data. Studies without metrics cannot contribute to the comparative analysis.
Inclusion	I5	Addresses at least one of: (a) ML/DL applied to GHG or flux prediction; (b) time series forecasting architectures with environmental or multivariate applicability; (c) agricultural/farm-scale digital twin systems; (d) GHG emission drivers in temperate livestock/grassland; (e) multi-source data fusion for ML pipelines	Directly maps to the five research questions defined in CWa-T2. Ensures all included evidence is actionable for the dissertation.
Inclusion	I6	Full-text accessible via UCL library access, open access, or institutional repository	Practical requirement for quality assessment. Studies behind paywalls with no alternative access route cannot be evaluated.
Exclusion	E1	Review or meta-analysis articles without novel primary data or modelling results, EXCEPT reviews of agricultural digital twin maturity (e.g. Purcell & Neubauer 2023) which directly address Research Question 3	Avoids double-counting evidence. Systematic reviews of DT maturity are explicitly included as they constitute primary evidence for the state-of-field assessment.
Exclusion	E2	Studies focused exclusively on crop yield, soil nutrient, or water quality without a GHG or flux prediction component	Ensures disciplinary focus. Yield prediction is a distinct problem domain with different drivers, data structures, and model requirements.
Exclusion	E3	Animal-scale enteric CH ₄ studies using only individual breath analysis without any connection to ecosystem-scale measurement or flux aggregation	Maintains the critical scale distinction between animal-scale (GreenFeed/chamber) and ecosystem-scale (EC) CH ₄ modelling. Animal-scale studies are included only where they inform the ecosystem measurement or contribute driver analysis.
Exclusion	E4	Studies applying ML to purely synthetic or simulated data without real sensor validation	Prioritises empirically grounded evidence. The dissertation uses real NWFP EC data; purely synthetic studies cannot provide reliable precedent for expected model performance.

APPENDIX E. Quality Assessment Criteria (CWA-T6)

Table 5: Quality Assessment

Quality Criterion	Evaluation Questions
Q-A: Study Design & Scope	1. Is the research question clearly stated, and does the study address at least one of the five dissertation research areas? 2. Is the study design appropriate for the claim being made (e.g. controlled benchmark vs. exploratory single-site study)? 3. Are the limitations of the study design explicitly acknowledged?
Q-B: Dataset Characteristics	4. Is the dataset clearly described, including size (number of samples/time steps), temporal resolution, and ecosystem or system type? 5. Is the dataset from a real, empirically measured system (not purely simulated)? 6. For EC flux studies: is the data from a site with managed livestock or temperate grassland
Q-C: ML Methodology	7. Are hyperparameter choices documented and justified (e.g. learning rate, sequence length, tree depth)? 8. Is the train-test split or cross-validation strategy clearly defined and appropriate for time series (i.e. avoids data leakage across time)? 9. Is at least one meaningful baseline model included for comparison? 10. Are feature selection or engineering decisions justified relative to domain knowledge?
Q-D: Evaluation & Metrics	11. Are quantitative evaluation metrics reported (RMSE, MAE, R^2 , NSE, or domain-equivalent)? 12. Are uncertainty estimates provided (confidence intervals, prediction intervals, or probabilistic outputs)? 13. Are results reported per test fold or across held-out periods (not aggregated over training data)? 14. Is statistical significance or variability of results assessed across runs or sites?
Q-E: Reproducibility	15. Is the code, dataset, or sufficient methodological detail provided to reproduce results? 16. Are data preprocessing steps (normalisation, gap-filling, quality control) explicitly described? 17. Are model architecture parameters and training procedures documented at reproducible detail?
Q-F: Relevance & Transferability	18. Does the study evaluate the model on data with comparable characteristics to NWFP EC data (half-hourly or sub-hourly, multi-sensor, outdoor environmental)? 19. Are findings generalisable beyond a single site or species, or explicitly scoped to site-specific conditions? 20. Does the study acknowledge domain-specific challenges (non-stationarity, missing data, seasonal variability) relevant to agricultural GHG forecasting?

APPENDIX F. Study Selection Process (Cwa-T7 & Cwa-T8)

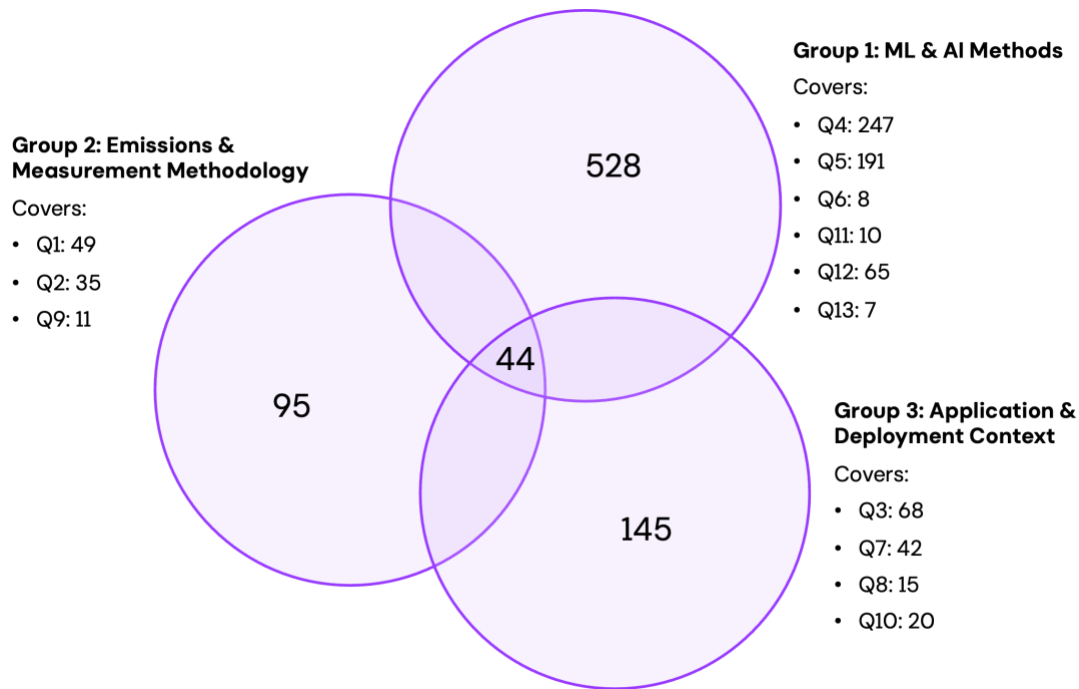


Figure 1. Venn diagram of results broken down by subjects.

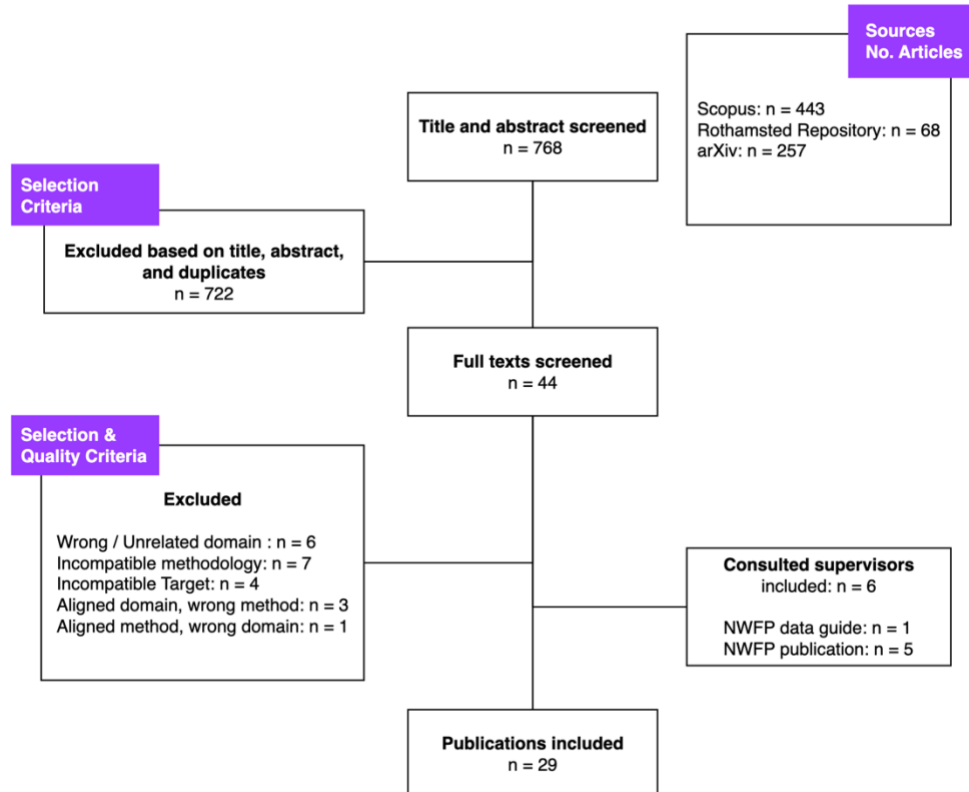


Figure 2. PRISMA selection flow chart. Final text: n = 29

APPENDIX G. Data Extraction Table

Table 6: Data Extraction Table

Study ID Year	Study Characteristics	Study Data	Study Methods	Study Results (Metrics)	Study Conclusions	AI Model Evaluation	Related RQ
AREA 1 — EC-ML GHG Prediction							
[7] Irvin et al. 2021	Landmark multi-site CH ₄ gap-filling benchmark. 17 FLUXNET-CH ₄ wetland sites; boreal to tropical biomes.	Half-hourly EC CH ₄ flux; FLUXNET-CH ₄ network; multiple site-years; soil temp, water table, met drivers.	MDS, LASSO, ANN, RF, XGBoost; artificial gap scenarios; cross-validation; uncertainty calibration method.	RF and XGBoost best across all sites. Soil temp most important predictor. ML uncertainty systematically underestimated. R ² ~0.7–0.8.	RF/XGBoost dominate EC CH ₄ gap-filling. UQ calibration essential. Open-source Python code released.	Strong: large multi-site benchmark; rigorous cross-validation; reproducible workflow; no managed grassland sites; no deep learning.	RQ1, RQ2
[8] Kim et al. 2020	EC CH ₄ gap-filling comparison; 5 wetland/rice paddy sites; temperate and tropical.	Half-hourly EC CH ₄ ; AmeriFlux and AsiaFlux sites; multiple meteorological and soil drivers; PCA-derived inputs.	ANN, RF, SVM vs MDS; PCA input reduction test; leave-one-site-out cross-validation.	RF outperforms ANN, SVM, MDS. Lagged variables improve CH ₄ more than CO ₂ . PCA degrades ML performance.	RF is robust baseline for EC CH ₄ . Feature engineering crucial; CH ₄ inherently harder to predict than CO ₂ .	Moderate-strong: no deep learning architectures; no UQ; no managed agricultural sites.	RQ1, RQ5
[9] Zhu et al. 2023a	EC gap-filling in challenging ecosystems including UK managed pastures and converted peatlands.	CO ₂ , water, energy, CH ₄ EC fluxes; European managed pastures; SE Asian peatlands; NA drylands; ERA5 reanalysis.	RFR vs MDS; gap length sensitivity (short/medium/long); ERA5 substitution test.	RFR outperforms MDS for gaps >12 days. ERA5 substitution validated. Key env-flux responses preserved in gap-filled data.	RFR recommended for managed pasture CH ₄ gap-filling; ERA5 viable backup when local sensors fail.	Strong: includes UK managed pasture; ERA5 validation directly applicable to NWFP pipeline.	RQ1, RQ5
[16] Kheradmand et al. 2023	ML for NEE prediction in grassland and cropland; 3 AmeriFlux sites; 7 years hourly data.	Hourly EC CO ₂ flux; grassland and cropland AmeriFlux sites; soil temp and met inputs.	ANN vs novel IOGAFS (genetic algorithm + Fourier series); normalised smoothed inputs.	IOGAFS: R ² =0.86 (CRO), 0.75 (GRA). ANN: R ² =0.88 (CRO), 0.81 (GRA). Both acceptable.	Novel GA-based model competitive with ANN for NEE; applicable where no flux tower available.	Moderate: CO ₂ /NEE only, not CH ₄ ; no RF/XGBoost baseline; no UQ; single novel model.	RQ1
[17] Bigaignon et al. 2020	N ₂ O gap-filling using ANN and linear interpolation; SW France cropland; 5-year automated chambers.	Daily N ₂ O flux; automated chambers; wheat-maize rotation; 2012–2016.	Linear interpolation alone vs ANN alone vs mixed ANN + linear; R ² and RMSE evaluation.	ANN alone: R ² =0.84, RMSE=12.4 gN/ha/d. Linear: R ² =0.68. Mixed improves low-flux periods.	Mixed ANN + linear recommended for N ₂ O gap-filling; ANN captures peaks, linear handles low-flux regimes.	Moderate: chambers not EC; N ₂ O not CH ₄ ; no tree-based baselines; cropland not grassland.	RQ1
[18] Goodrich et al. 2021	Improved EC N ₂ O gap-filling and uncertainty estimation; New Zealand dairy grazed peat grassland.	Half-hourly EC N ₂ O flux; NZ peat grassland; met and soil drivers; multi-year record.	Multiple gap-filling approaches; bootstrapped uncertainty estimation; comparison with REddyProc.	Improved UQ reduces uncertainty vs standard REddyProc. Bootstrapping provides realistic confidence intervals.	Standard gap-filling tools underestimate N ₂ O uncertainty; improved UQ essential for reliable budgets.	Moderate-strong: grazed pasture EC directly relevant; N ₂ O not CH ₄ ; provides UQ methodology.	RQ1, RQ2
[19] Jung et al. 2020	FLUXCOM: global upscaling of EC carbon fluxes using ML; systematic multi-method assessment.	FLUXNET global network; GPP and NEE; multiple ML methods; satellite remote sensing drivers.	Multiple ML methods (RF, ANN, SVM); multiple forcing datasets; cross-validation against inversions and DGVMs.	GPP spatial patterns consistent. NEE seasonal cycles match inversions. Interannual variability underestimated.	ML upscaling provides useful global constraints; tropical uncertainty highest; fertilisation not captured.	Strong: gold standard for EC upscaling methodology; CO ₂ /GPP not CH ₄ ; global not farm-scale.	RQ1, RQ5
[20] Liang et al. 2020	Global grassland NEE upscaling using model tree ensemble; 44 flux tower sites; 1982–2011.	Monthly NEE from 44 grassland towers (1,457 site-months); radiation, temperature, LAI, fPAR drivers.	Model tree ensemble (MTE-GM) with management activities; NSE and R ² evaluation; cross-validation.	NSE=0.90 (among-site), 0.86 (seasonal). Mean global grassland NEE: 72±4 gC/m ² /yr (small source).	Temperate grasslands act as carbon sinks; management activities significant; MTE-GM generalises well.	Moderate: CO ₂ /NEE not CH ₄ ; global scale not farm scale; no deep learning; provides grassland ML precedent.	RQ1

COMP0190: Project Preparation Module
Structured Literature Review Coursework

Study ID Year	Study Characteristics	Study Data	Study Methods	Study Results (Metrics)	Study Conclusions	AI Model Evaluation	Related RQ
[21] McNicol et al. 2023	UpCH4: RF upscaling of FLUXNET-CH4 wetland CH4 emissions; 43 sites; 2001-2018.	EC CH4 from 43 FLUXNET-CH4 wetland sites (119 site-years); 6 predictors; global gridded output.	RF upscaling model; NSE evaluation; comparison with land surface models and atmospheric inversions.	NSE ~0.52-0.63 extra-tropically. Global wetland CH4: 146±43 TgCH4/yr. Tropical uncertainty high.	RF upscaling viable for wetland CH4; tropical regions remain main uncertainty source.	Strong: EC CH4 upscaling benchmark; wetland not agricultural grassland; no deep learning.	RQ1
[22] Ouyang et al. 2023	Paddy rice CH4 upscaling across Monsoon Asia using RF and MODIS; 23 global EC sites.	EC CH4 from 23 global rice paddy EC sites; MODIS remote sensing; 2001-2015; 5 km gridded output.	RF model; NSE evaluation; SHAP feature importance; gridded upscaling.	NSE=0.59 (8-day fluxes), 0.69 (site means). Land surface temp, biomass, water availability most important.	RF viable for rice CH4 upscaling; declining emissions 2007-2015; temperature and vegetation key drivers.	Moderate-strong: CH4 EC + ML; rice paddy not managed grassland; SHAP interpretability relevant to RQ4.	RQ1, RQ4
[23] Ma et al. 2025	Combined MDS + ML for CH4 EC gap-filling in rice paddy; improves performance vs either alone.	Half-hourly EC CH4; paddy fields; meteorological and soil drivers; multiple gap scenarios.	MDS alone vs ML alone vs combined MDS+ML hybrid; multiple gap lengths; R2 and RMSE.	Combined MDS+ML outperforms either method alone across gap lengths. Improved R2 and RMSE.	Hybrid gap-filling outperforms single-method approaches for CH4; ensemble approach recommended.	Moderate: recent 2025 paper; rice paddy not grassland; hybrid methodology directly applicable to NWFP.	RQ1
[13] Sharma et al. 2026	Coupled ML-ecosystem ensemble for N2O prediction in US croplands; PNAS 2026.	N2O fluxes from US cropland network; ecosystem model outputs as ML features; multi-site.	ML coupled with ecosystem model outputs; ensemble approach; comparison with process-based alone.	Coupled ML-ecosystem substantially improves N2O predictions vs process-based alone.	Hybrid process-ML models outperform either approach; ecosystem model outputs are valuable ML features.	Strong: novel hybrid methodology directly relevant to SPACSYS+ML approach; N2O not CH4; cropland.	RQ1
AREA 2 — Time Series Forecasting							
[10] Zeng et al. 2023	Critical benchmark: DLinear/NLinear vs transformer architectures for LTSF; AAAI 2023.	9 real-world benchmark datasets (ETT, Weather, ECL, Traffic, Exchange); standard LTSF benchmarks.	DLinear (decomposition + linear) and NLinear vs Informer, Autoformer, FEDformer; 9 datasets.	DLinear/NLinear outperform all transformers on all 9 benchmarks, often by large margins.	Permutation-invariant attention loses temporal ordering; simple models often sufficient for LTSF.	Strong: rigorous benchmark; directly challenges DL assumption; motivates strong baselines in dissertation.	RQ1
AREA 3 — Agricultural Digital Twins							
[4] Fakeye et al. 2024	NWFP digital twin framework; three-tier architecture; MODELS Companion 2024.	NWFP Devon UK; up to 15-minute sensor data; three demonstrator DTs under development.	Three-tier architecture (client/analysis/storage); Kubernetes orchestration; PBMs + DDMs + hybrid ensemble.	No metrics. Framework described; demonstrators under development; LandscapeDNDC, RothC_Py integrated.	NWFP DT architecture defined; CH4 forecasting module is identified gap; ensemble modelling strategy.	Moderate: conference paper; framework not fully validated; directly describes dissertation infrastructure.	RQ3
[5] Purcell & Neubauer 2023a	PRISMA systematic review of agricultural DTs; state-of-the-art; what-if simulation gap identified.	DT applications literature 2018-2022; PRISMA methodology; diverse agricultural applications.	PRISMA systematic review; DT classification by capability; maturity assessment.	What-if simulation absent from most systems; most DTs conceptual; whole-farm scale rare.	What-if simulation is the central unmet DT capability; most systems are digital shadows.	Strong: PRISMA methodology; directly identifies dissertation gap; key definitional reference.	RQ3
[6] Purcell et al. 2023b	Exploratory review of DTs for sustainable agriculture; food security and climate change framing.	DT literature review; sustainability dimensions; social-ecological risks.	Exploratory review; sustainability roadmap; risk assessment.	DTs uniquely positioned for sustainability; what-if scenarios key capability; negative effects must be assessed.	DT sustainability framing; scenario simulation validates dissertation objective; warns negative effects.	Moderate-strong: complements 2023a; less rigorous methodology; sustainability framing relevant.	RQ3
[24] Mishra & Sharma 2023	IoT and Digital Twins in Livestock Farming (DTLF); PLF sensor integration; animal	Review of DT applications in livestock; GPS, RFID, AI, ML technologies;	Conceptual review; technology assessment; performance claims for DTLF system.	92% disease prediction improvement claimed; 94% accuracy for	IoT-DT integration in livestock farming; PLF sensors provide DT	Weak-moderate: performance claims not rigorously validated;	RQ3

COMP0190: Project Preparation Module
Structured Literature Review Coursework

Study ID Year	Study Characteristics	Study Data	Study Methods	Study Results (Metrics)	Study Conclusions	AI Model Evaluation	Related RQ
	welfare monitoring.	animal health monitoring.		temperature and humidity monitoring.	foundation; high variability challenges.	livestock context relevant to NWFP.	
AREA 4 — Emissions Drivers							
[25] Cardenas et al. 2022	CO ₂ EC flux at NWFP; three pastures (PP, HS, HSC); 2017-2018; Prof. Harris co-author.	Half-hourly EC CO ₂ ; NWFP Devon; PP, HS, HSC pastures; 2-year measurement record.	EC gap-filling; REddyProc; regression modelling; pasture type comparison.	PP: sink in 2017 (-5.40 tCO ₂ /ha/yr) to source in 2018 (+6.17). Season and pasture type significant.	Inter-annual CO ₂ variability large at NWFP; pasture type and season interact; management drives flux.	Strong: directly at NWFP; same EC infrastructure as dissertation; validates site variability patterns.	RQ4, RQ5
[26] Feigenwinter et al. 2023	N ₂ O and CH ₄ EC in intensively managed temperate grassland; Swiss Chamau site; N ₂ O mitigation experiment.	Half-hourly EC N ₂ O and CH ₄ ; Chamau Switzerland; fertilised vs clover parcel; 2019-2020.	RF gap-filling; driver analysis; WFPS, precipitation lagged variables; management event analysis.	RF gap-filling captures emission peaks. Lagged precip and WFPS most important. Clover reduces N ₂ O 36-54%.	RF viable for N ₂ O and CH ₄ EC in managed grassland; management and biological drivers interact.	Strong: managed temperate grassland EC; closest ecosystem match to NWFP; RF gap-filling methodology.	RQ4
[27] Barczyk et al. 2024	EC N ₂ O driver partitioning from grazed and fertilised Swiss pasture; Agric. For. Meteorol. 2024.	EC N ₂ O fluxes; Swiss grazed fertilised pasture; management events (grazing, fertilisation); 3-year record.	EC flux partitioning; driver analysis; management event attribution.	Management events explain large fraction of N ₂ O variability; grazing and fertilisation are dominant drivers.	Management events critical for N ₂ O prediction; event-aware modelling required.	Strong: EC N ₂ O on grazed pasture; management event analysis directly relevant to NWFP scenario design.	RQ4
[28] Jones et al. 2017	Full GHG budget of grazed, cut and fertilised temperate grassland; SE Scotland; 9 years.	EC CO ₂ (NEE); N ₂ O and CH ₄ estimates; N and C stocks; SE Scotland; 2002-2010; 9 years.	Budget calculation; all imports/exports; modelled vs measured GHG components.	NEE uptake 218±155 gC/m ² /yr. GHG sink reduced 54% by CH ₄ +N ₂ O. Enteric CH ₄ dominant CH ₄ source.	Enteric fermentation dominates CH ₄ budget in grazed grassland; N ₂ O and CH ₄ substantially offset CO ₂ sink.	Strong: comprehensive GHG budget; temperate managed grassland; 9-year record; directly informs RQ4.	RQ4
[29] Knox et al. 2016	Biophysical controls on interannual EC CO ₂ and CH ₄ variability; California rice paddy; 6.5 years.	Half-hourly EC CH ₄ and CO ₂ ; California rice paddy; 6.5 years; wavelet analysis.	Wavelet analysis; forward stepwise linear models; neural networks; interannual variability assessment.	GEP drives diel CH ₄ pattern; soil temp modulates amplitude and interannual variability. NN > linear.	Soil temperature explains most interannual CH ₄ variability; NN outperforms linear for nonlinear CH ₄ response.	Moderate: rice paddy not managed grassland; long record; biophysical driver analysis relevant to RQ4.	RQ4
[30] Fuchs et al. 2020	Multimodal evaluation of N ₂ O in intensively managed Swiss grassland; DayCent, PaSim, APSIM.	Field N ₂ O measurements; Swiss managed grassland; control vs clover treatment; 4 years.	DayCent (2 variants), PaSim, APSIM (2 variants); ensemble average vs IPCC-Swiss benchmark.	Ensemble reduces error 41% vs IPCC method. DayCent best annual; APSIM best daily. Ensemble overestimates mitigation.	Multimodal ensembles outperform IPCC Tier methods; process model limitations in N ₂ O remain.	Moderate: process models not ML; N ₂ O not CH ₄ ; Swiss managed grassland; ensemble methodology relevant.	RQ4
[31] da Silva Pontes et al. 2018	CH ₄ from beef heifers in temperate integrated crop-livestock systems; Brazil; 3 years.	GreenFeed-type CH ₄ measurements; beef heifers; 3 years; CL vs CLT systems; N fertilisation levels.	CH ₄ per animal, per LW, per kg ADG, per ha; treatment comparison; statistical analysis.	Mean CH ₄ : 163±9 g/day/heifer. Tree shading reduces ADG 32% but CH ₄ /LW unchanged. N level significant.	Pasture management (shade, N) affects area-based CH ₄ but not per-LW emissions; forage quality key.	Moderate: South American tropical system not UK temperate; animal-scale measurements; CH ₄ drivers relevant.	RQ4
[32] Feron et al. 2024	Climate-driven CH ₄ flux trends in boreal/temperate wetlands; FLUXNET-CH ₄ ; 1981-2018 hindcast.	EC CH ₄ from 22 FLUXNET-CH ₄ wetland sites; gradient boosting hindcast; ERA5 reanalysis drivers.	Gradient boosting model; meteorological reanalysis hindcast; trend analysis; extreme flux analysis.	50% of sites show significant CH ₄ increases. Fall season most affected. Extreme flux days increasing.	Temperature-driven CH ₄ increases extending emission seasons; extreme flux days now 10-40% of seasonal totals.	Moderate: wetland not managed grassland; provides climate-CH ₄ trend context; gradient boosting methodology.	RQ4
[12] Buzacott et al. 2024	CH ₄ drivers and annual totals from	EC CH ₄ from 16 Dutch peatland	ML gap-filling; SHAP feature importance; annual	Soil temperature most important	Temperature and GWL dominate	Moderate-strong: EC CH ₄ + SHAP	RQ4

COMP0190: Project Preparation Module
Structured Literature Review Coursework

Study ID Year	Study Characteristics	Study Data	Study Methods	Study Results (Metrics)	Study Conclusions	AI Model Evaluation	Related RQ
	Dutch peatlands; 16 EC sites; ML gap-filling with SHAP.	sites; 6 land use classes; GWL and temperature as main drivers.	budget estimation; land use comparison.	predictor. GWL dominant driver. Annual totals 90-632 kgCH4/ha/yr by land use.	CH4 in peatlands; SHAP reveals multi-driver dependency.	interpretability; peatland not managed grassland; driver analysis relevant.	
AREA 5 — Multi-source Data & UQ							
[15] Oulaid et al. 2025	Geographically weighted quantile ML for soil moisture UQ at NWFP; 15 SM stations; 6 land uses.	Daily soil moisture; NWFP Devon; 15 stations; 6 land uses; satellite radar backscatter + ERA5 + topography.	GWQML with Gaussian/Tricube kernels; quantile prediction; Leave-One-Land-Use-Out validation; Moran's I.	R2 up to 0.85; PICP up to 0.9; GWQML outperforms global QML; reduced spatial autocorrelation in residuals.	Spatial UQ framework applicable to NWFP; quantile ML provides calibrated uncertainty; spatial weighting improves coverage.	Strong: directly at NWFP; same site as dissertation; UQ methodology; soil moisture not CH4 flux.	RQ2, RQ5
[33] Zhu et al. 2023b	EC flux extrapolation to field scale at 30m resolution; European managed grazing pastures; NWFP-adjacent.	EC CO2/CH4 fluxes; 206 FLUXNET2015 sites; 3 European managed pastures; Sentinel-2; 30m resolution.	ML extrapolation; satellite resolution comparison; R2 and annual bias; livestock grazing resistance test.	R2 ~0.7; annual bias <2 Mg/ha/yr for carbon. Sentinel-2 best for EU pastures. ML resistant to livestock grazing.	Field-scale EC flux estimation viable at 30m; Sentinel-2 captures land-use variability; NWFP pastures included.	Strong: NWFP pastures included; managed livestock grassland; field-scale methodology; data fusion relevant.	RQ5
[11] Partridge et al. 2024	ML for NWFP cattle CH4 prediction; GreenFeed + farm variables; ICCA 2024 conference paper.	NWFP GreenFeed CH4 measurements; farm environmental and biological variables;	Gradient Boosting, RF; XAI/SHAP feature importance; r and RMSE evaluation.	Gradient Boosting best (g/day: r=0.619, RMSE=51.8; g/kg LWG: r=0.562, RMSE=65.9). XAI applied.	Gradient Boosting best for NWFP cattle CH4; r~0.6 modest; LSTM/TFT temporal architectures untested.	Moderate-strong: directly at NWFP; same GreenFeed data; most overlapping prior work; animal-scale not EC.	RQ1, RQ4